WORKING PAPER · FOR DISCUSSION

The Golden Third

A static costing and distributional model of a shared tax and social-protection framework for a group of European states

Initiative 28:28

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Abstract

We describe a proposal under which a group of states adopts common rules for income taxation and social protection, and we document a transparent, static model of its costs and distributional effects. The proposal pursues four objectives: tax fairness across labour, capital and automation; fairness between generations; coping with climate change; and coping with the digital transformation of work. Income of every kind is taxed in steps of 0, 11, 19, 25 and 33 per cent tied to each country's basic need; social contributions are abolished and employers must pass at least 80 per cent of the employer contributions that disappear into wages; a tax-financed safety net covers health care, long-term care and basic needs; every adult receives a monthly transformation dividend; 10 per cent of gross wages go into a personal, locked pension account; and public debt and unfunded pension promises are moved into a common fund repaid by 2098. For 2028, in purchasing power of 2026, a single employee at the average wage gains between 276 € and 726 € per month in Germany, France, Italy and Spain and 1,404 zł in Poland. The Gini coefficient falls in all five countries, by 6.2 points at most. The common fund requires 5.17 per cent of GDP a year, a surcharge of 15.5 per cent on all taxes at an interest rate of 2.45 per cent, and peaks at 148.1 per cent of GDP in 2055. Losers are German pensioner households (only 32.2 per cent better off), single earners in France at half the average wage, and one-earner couples at twice the average wage. The model has no behavioural responses; the main uncertainties are the pass-through of employer contributions, the fund interest rate and the tax base of company profits taxed where customers are. Measured against its objectives, the model supports the claims on distribution and on the robustness of revenue to a shift from labour to capital income; it shows tensions between climate success and the size of the dividend, and between legacy repayment and today's pensioners; and it cannot yet show effects on emissions.

Keywords: tax reform, social protection, basic income, transformation dividend, payroll tax incidence, public debt, Europe. **JEL:** H20, H24, H55, H63, I38.

1 Introduction

Most European states finance social protection to a large extent through contributions levied on wages, combine them with progressive income taxes and an array of means-tested benefits, and carry public debt and unfunded pension promises from the past. The resulting systems are complex, and the marginal burden on additional earnings is high in parts of the income distribution. The proposal examined here – named *The Golden Third* after its upper tax rate of one third – replaces this structure with a small number of common rules that a group of states could adopt jointly.

This paper does not derive an optimal tax system. It is a costing and distributional accounting exercise: given the rules, what does the system cost, who gains and who loses compared with the law of 2026, and how sensitive are these results to the main assumptions? All figures come from one calculation chain of 73 stages based on official sta-

tistics. Every change to the model is recorded in a log, and every error found is numbered as a correction (85 so far). The chain was recalculated in three internal reviews, carried out with a different AI model from the one used to build the calculation; the last review covered model version v4.42 (section 10).

Section 2 states the objectives, section 3 the rules, section 4 the model and data, section 5 the results, section 6 the sensitivity to key assumptions. Section 7 assesses the results against the objectives, including the tensions between them. Section 8 summarises behavioural evidence that the static model does not capture, section 9 limitations and open questions, section 10 the review history.

2 Objectives

The initiative states four objectives. They are normative choices of the initiative; this paper does not argue for them but asks whether the rules are consistent with them and what the model can show.

Tax fairness. Today labour carries most of the burden: in Germany the tax wedge on an average wage is about 49.3 % of labour costs (OECD 2026), while capital income is taxed at a flat rate of about 26 %. Social contributions fall on human work only, so replacing work by machines saves contributions. The proposal treats labour, capital and automation alike: no income is taxed at more than one third, without deductions, and company profits – explicitly including digital platforms – are taxed where the customers are, because profits can be shifted but customers cannot. Fairness is also vertical: a floor at basic need and a dividend for everyone.

Intergenerational fairness. Public debt and unfunded pension promises are commitments made by earlier generations that today's systems pass on to later ones. The proposal makes them explicit, moves them into a common fund and repays them by 2098. Pension promises above basic need are reduced, so that today's pensioners carry a share; basic need is guaranteed. New saving for old age is funded on an own account rather than promised by later contributors. The dividend is justified as a share for everyone in what earlier generations built – knowledge, technology, infrastructure.

Climate change. The proposal relies on economic incentives rather than on control of individual products and firms. A carbon price and a special VAT rate of 33 % on harmful goods make harmful consumption more expensive; their revenue is returned to everyone through the dividend, so that households that consume little gain. VAT becomes a label with five classes. Suppliers move their products into lower classes by certification at approved bodies under state supervision: the state supervises the certifiers, not each product. The intended result is a competition for sustainability in which a lower rate is a price advantage that suppliers earn. The dividend shrinks as emissions fall; the initiative regards this as intended.

Digital transformation of work. If more output is produced by machines and software and less by paid work, systems financed from wages lose their base and tie protection to employment that becomes less stable. The proposal finances social protection from all income, makes the net and the dividend independent of employment, keeps the pension account portable across jobs, and taxes platforms where their customers are. The dividend is meant to give everyone a share of what automation and digitalisation bring in.

3 The proposal

Income tax	Rates 0 / 11 / 19 / 25 / 33 % in brackets defined as multiples of each country's basic need. Gross income without deductions; individual taxation (no splitting or family quotient). Capital income, profits and inheritance are taxed at most at 33 %; company profits are taxed where customers are, including digital platforms.
Contributions	Social contributions are abolished. Employers are legally obliged to pass at least 80 % of the employer contributions to social security that disappear into gross wages, up to six times basic need; a fallback rule applies if contracts are not adjusted by the start date.

Strong net	Health care, long-term care (partial), minimum pension, basic needs in disability and unemployment, financed from general taxes. A floor equal to basic need times an equivalence scale (0.5 for each further adult, 0.55 per child) is guaranteed; own earnings count by half. In Germany the floor is 1,029 € per month for a single person (2028, purchasing power 2026). Low earners are topped up on an application that the authority pre-fills.
Dividend	Monthly payment to every adult, 80 % for children, without application; financed from steering taxes (carbon and other) and from gains of digitalisation. It falls as emissions fall.
Own route	Compulsory saving of 10 % of gross wages into a personal account, locked until retirement age, with few investment options and a default fund.
VAT label	Rates 5 / 11 / 17 / 25 %, special rate 33 % only for harmful goods. Suppliers obtain lower rates through certification by supervised bodies.
Legacy	Public debt and unfunded pension promises above the dividend move into a common fund of the participating states, financed by a surcharge on all taxes. Existing pension promises above basic need are reduced by 15 %; own savings are not reduced (pension rule E). The effective reduction of pension payments differs by state, from 1.9 % to 12.1 % (6.95 % in Germany).

4 Model and data

Scope. The calculations cover a possibility space of 30 European states with 465.6 million inhabitants and a GDP of 19,515 billion euro (2024). Ten states are calculated in detail; the common fund of 30 states is the decision of the ten scaled by the ratio 30/10 from the possibility space (assumption). Household results are computed for five reference countries: Germany, France, Italy, Spain and Poland.

Revenue and expenditure. Tax bases (wages, profits, consumption, wealth, inheritance) and current expenditure come from Eurostat national and government accounts (gov_10a_exp, gov_10a_taxag, nama_10_gdp); unfunded pension claims from the supplementary pension tables (nasa_10_pens1, table 29); debt from gov_10dd_edpt1. The tax base of company profits taxed where customers are is the largest position that is not yet documented against an independent source.

Households today. Each reference country has a simplified model of 2026 law: income tax, employee and employer contributions and cash benefits including minimum income schemes (Germany: Bürgergeld approximation with housing and child supplements; France: APL and prime d'activité; Italy: Assegno di Inclusione; Spain: Ingreso Mínimo Vital and child supplement; Poland: zasiłek okresowy). Parameters come partly from official sources and partly from secondary sources; the parameter files mark each block as official, secondary or assumption. The models were checked against the OECD tax wedge: 7 of 10 household types lie within ±2 percentage points (Table 3).

Germany. A synthetic microsimulation, calibrated to official totals, covers employees, civil servants, self-employed, minimum-income recipients and pensioners. Validation against 2025: gross wages −2.8 %, wage tax −5.4 %. Pensioner incomes come from EU-SILC 2025 and the German old-age income survey (ASID 2023), both as published aggregates. No micro data are used.

Distribution. The Gini coefficient for 2028 is the official value for 2025 (EU-SILC) plus the change computed on the published decile distribution. VAT burdens by income come from the German household budget survey (EVS 2023) and, for France, from Eurostat's Household Budget Survey 2020; Italy, Spain and Poland use a factor of 1 so far.

Conventions. Amounts for 2028 are in purchasing power of 2026. The model is static: labour supply, saving, prices beyond a one-off level effect and migration do not respond. Pass-through of employer contributions, take-up, avoidance and VAT incidence enter as parameters or bands. Take-up of the floor is assumed to be complete.

5 Results

5.1 Households in five countries

Table 1 shows the change in disposable income for three household types at half, once and twice the average wage. In Germany a single employee at the average wage gains +726 € per month; at a gross wage of € 42,000, about two thirds of the average wage, the gain is +611 € for a single person, +610 € for a single parent and +674 € for a one-earner couple with two children.

Table 1: Change in monthly disposable income, 2028

Country	Household	½ × AW	1 × AW	2 × AW	Better off
Germany	single	+586 €	+726 €	+710 €	98 %
	single parent, 1 child	+588 €	+721 €	+707 €	98 %
	couple, one earner, 2 children	+561 €	+688 €	+78 €	98 %
France	single	−148 €	+276 €	+459 €	73 %
	single parent, 1 child	+205 €	+228 €	+253 €	73 %
	couple, one earner, 2 children	+760 €	+380 €	0 €	98 %
Italy	single	+304 €	+412 €	+766 €	100 %
	single parent, 1 child	+413 €	+416 €	+824 €	100 %
	couple, one earner, 2 children	+1,054 €	+619 €	+1,021 €	100 %
Spain	single	+217 €	+321 €	+336 €	98 %
	single parent, 1 child	+477 €	+317 €	+401 €	100 %
	couple, one earner, 2 children	+1,233 €	+662 €	+701 €	100 %
Poland	single	+1,406 zł	+1,404 zł	+1,554 zł	100 %
	single parent, 1 child	+1,971 zł	+909 zł	+338 zł	98 %
	couple, one earner, 2 children	+4,029 zł	+2,246 zł	+722 zł	98 %

Change in monthly disposable income, 2028, purchasing power 2026; Poland in złoty. AW: average wage, OECD Taxing Wages 2024 (Germany 63,288 €; France 44,968 €; Italy 35,616 €; Spain 31,698 €; Poland 96,421 zł). Better off: share of employees of this household type across the wage distribution; the German wage distribution is scaled to each country's average wage (assumption). Each country carries its own legacy surcharge after netting in the common fund.

Three patterns stand out. First, one-earner couples with higher wages lose part of today's joint taxation (German splitting, French quotient familial), because every adult is taxed individually and receives an own dividend; around twice the average wage the gain is close to zero in Germany (+78 €) and France (0 €). Second, single earners in France at half the average wage lose 148 € per month: today's in-work benefit and family allowances are replaced by the dividend and the floor, and employer contributions near the minimum wage are already largely waived today, so little is passed on. Third, in Italy, Spain and Poland, where minimum income schemes reach fewer households today, the floor and the dividend raise low incomes more strongly.

Table 2: Inequality and legacy surcharge

Country	Gini 2025 (official)	Gini 2028 (model)	Change	Legacy surcharge 2028
Germany	30.1	26.1	−4.0	15.0 %
France	30.4	29.0	−1.4	21.3 %
Italy	31.0	26.9	−4.1	22.3 %
Spain	30.8	25.0	−5.8	18.3 %
Poland	24.9	18.7	−6.2	13.5 %

Gini coefficient of equivalised disposable income (0–100). 2025: Eurostat ilc_di12 (EU-SILC 2025). 2028: official value plus the change computed in the model from the decile distribution (Eurostat ilc_di01); not a microsimulation except for Germany.

Table 3: Validation of the country models against the OECD tax wedge

Country	Single: model	OECD	Diff. (pp)	Couple, 2 ch.: model	OECD	Diff. (pp)
Germany	48.9	49.3	−0.4	34.7	34.9	−0.2
France	46.0	47.2	−1.2	33.5	39.1	−5.6
Italy	44.4	45.8	−1.4	33.0	34.3	−1.3
Spain	42.4	41.4	+1.0	37.8	36.8	+1.0
Poland	40.3	35.0	+5.3	20.7	14.2	+6.5

Tax wedge in % of labour costs at the average wage. Model: 2026 law; OECD: Taxing Wages 2026 (2025 data).

Poland: the OECD counts neither accident insurance nor the labour and guarantee funds on the employer side, and treats the health contribution differently (open). France, couple: the model pays the prime d'activité at this wage level, consistent with ministry examples for the 2026 reform; the OECD apparently does not (to be checked).

5.2 Germany: all households and pensioners

In the German microsimulation 85.1 % of working-age households and 69.3 % of all households are better off. By status of the main earner: full-time 99.1 %, part-time 99.9 %, self-employed 77.7 %, civil servants 56.4 %, mini-job 49.1 %, households without earnings 18.0 %. Under pension rule E, 32.2 % of pensioner households are better off; the mean change is −121 € per month, from +24 € in the lowest to −478 € in the highest fifth. Pensioners lose because their contributions to health and long-term care disappear while pension promises above basic need are reduced, and because the dividend does not fully replace these amounts in the upper part of the distribution.

5.3 Public finances

Table 4 shows the strong net and the dividend for Germany. The dividend amounts to 354 € per adult and 283 € per child and month in 2028 (Table 4 shows the dividend in per cent of GDP). Its share of GDP falls from 7.25 % in 2028 to 3.40 % in 2060 because steering revenue falls with emissions. The total burden of taxes and fund is 38.7 % of GDP in 2028 against 40.76 % today (Eurostat 2024); including compulsory saving, which goes to an own account, it is 42.8 %.

Table 4: Strong net and dividend, Germany

Germany, % of GDP	**2028**	**2060**
Health care (basic coverage)	7.36	7.63
Long-term care (partial)	0.85	1.41
Minimum pension (67+)	0.00	0.53
Disability (basic need)	0.30	0.30
Unemployment (basic need)	0.42	0.42
Orphans	0.05	0.05
Hardship (housing, special needs)	0.65	0.65
Strong net, total	**9.63**	**10.98**
Transformation dividend	7.25	3.40
of which financed by steering taxes	3.14	0.67

netz33.json (decision path). Quantities (pensioners, disabled, unemployed) and the reduction of the benefit catalogue are assumptions.

The common fund takes over public debt and pension claims above the dividend (on average 9.42 % of GDP in 2028). At a fund interest rate of 2.45 % – the euro-area average of 2006–2025 – it requires 5.17 % of GDP a year, fi-

nanced by a surcharge of 15.5 % on all taxes. Its debt peaks at 148.1 % of GDP in 2055 and is repaid in 2098. Netting by state means that states with larger legacies carry a higher surcharge until 2058 (Table 5).

Table 5: Contribution and surcharge by state after netting

State	Contribution 2028 (% GDP)	Surcharge 2028 (%)
Germany	5.00	15.0
France	7.12	21.3
Italy	7.44	22.3
Spain	6.14	18.3
Poland	4.50	13.5

After netting in the common fund; converging to the common rate (15.5 %) by 2058.

5.4 VAT label and price level

The additional VAT burden is moderately progressive: relative to the mean it ranges from 0.87 in the lowest to 1.07 in the highest German income class, because rent, health, education and insurance are exempt (Table 6). The one-off price-level effect in 2028 is 2.47 % mechanically and 1.2 to 2.1 % with partial pass-through on taxable consumption only. VAT increases are passed on more strongly than cuts (Benzarti et al. 2020), so the upper end of the band is more likely; an asymmetric variant is not yet computed.

Table 6: VAT label burden by income

Germany: household net income per month	Factor
<1000 €	0.87
1000-1300 €	0.91
1300-1500 €	0.92
1500-2000 €	0.96
2000-2600 €	1.00
2600-3600 €	1.01
3600-5000 €	1.03
≥ 5000 €	1.07

EVS 2023. Factor: additional VAT burden relative to the mean, as share of consumption.

France: income quintile	Factor
Q1	0.90
Q2	0.91
Q3	1.00
Q4	1.06
Q5	1.12

Eurostat HBS 2020 (hbs_str_t223, values flagged as estimated).

5.5 Marginal burden

Comparing new taxes with today's taxes alone would overstate the gain, because health, care and pensions are financed by contributions today and by taxes in the new system. Table 7 therefore adjusts today's burden for the part of the pension contribution that buys an own entitlement. At € 60,000 the burden on an additional euro falls from 48.8 % today to 40.0 % in 2028.

Table 7: Marginal burden, Germany, single

Gross wage	Today	Today, adjusted (0.5 / 1)	New 2028	New incl. compulsory saving
20,000 €	35.8	31.1 / 26.5	32.0	40.5
42,000 €	43.8	39.1 / 34.5	40.3	48.9
60,000 €	48.8	44.1 / 39.5	40.0	48.5
100,000 €	53.1	48.5 / 43.8	39.1	49.1
250,000 €	44.3	44.3 / 44.3	40.0	50.0

Germany, single, % per additional euro of gross wage. Adjusted: the part of the pension contribution that creates an own entitlement is treated as saving (equivalence 0.5 or 1, assumption). Employer side and means-tested benefits not included.

6 Sensitivity

Pass-through of employer contributions. This is the most important single assumption. Evidence from Sweden shows that employers did not pass a cut in payroll taxes for young workers into their wages individually (Saez, Schoefer and Seim 2019); the proposal therefore makes pass-through a legal obligation with a fallback rule. Without any pass-through, the German single at € 42,000 still gains +258 €, but in France only 38 % and in Spain 53 % of single employees would be better off (Table 8).

Table 8: Pass-through of employer contributions

Pass-through	DE single € 42,000	Better off DE	Better off FR	Better off IT	Better off ES	Better off PL
0 %	+258 €	84 %	38 %	98 %	53 %	98 %
50 %	+479 €	84 %	71 %	100 %	98 %	100 %
80 %	+611 €	98 %	73 %	100 %	98 %	100 %
100 %	+699 €	100 %	73 %	100 %	100 %	100 %

Share of the employer contributions that disappear which is passed into gross wages; the remainder stays with the employer. Better off: single employees. 80 % is the adopted value (legal obligation with fallback rule).

Fund interest and growth. At a fixed rate of 3.5 %, close to German federal bond yields in October 2026, the surcharge rises to 18.7 % and the German single at € 42,000 gains +570 € instead of +611 € (Table 9). Sustainability depends on the gap between interest and growth: with low nominal growth and a contribution fixed at the adopted value, debt would not be repaid by 2098 (Table 10). Holding the pension claims Z constant in per cent of GDP across growth paths is a simplification.

Table 9: Fund interest rate

Fund interest	Contribution (% GDP)	Surcharge (%)	DE single, € 42,000
fixed 2.45 %	5.17	15.5	+611 €
fixed 3.0 %	5.74	17.1	+590 €
fixed 3.5 %	6.27	18.7	+570 €
fixed 4.0 %	6.82	20.4	+548 €
transition 2.22 → 3.5 %	6.20	18.5	+572 €
transition 2.22 → 4.0 %	6.70	20.0	+553 €

Nominal growth 3.0 %. Market yields October 2026 (secondary sources): German federal bonds 3.46 %, EU bonds 3.95 %.

Table 10: Interest and nominal growth

Interest	Nominal growth	Contribution (% GDP)	Surcharge (%)	Debt 2098 at fixed contribution (% GDP)
2.45 %	2.0 %	6.22	18.6	87.4
3.00 %	2.0 %	6.83	20.4	168.8

Interest	Nominal growth	Contribution (% GDP)	Surcharge (%)	Debt 2098 at fixed contribution (% GDP)
3.50 %	2.0 %	7.40	22.1	274.3
4.00 %	2.0 %	7.98	23.9	421.1
2.45 %	2.5 %	5.68	17.0	35.9
3.00 %	2.5 %	6.27	18.8	93.3
3.50 %	2.5 %	6.83	20.4	168.0
4.00 %	2.5 %	7.40	22.1	272.5
2.45 %	3.0 %	5.17	15.5	0.1
3.00 %	3.0 %	5.74	17.1	40.2
3.50 %	3.0 %	6.27	18.7	92.9
4.00 %	3.0 %	6.82	20.4	167.1

Contribution and surcharge needed for repayment by 2098; last column: remaining debt (ten states) if the contribution stays at the adopted value.

Avoidance at the top. Revenue from inheritance (1.16 % of GDP) and large wealth (0.97 % of GDP after avoidance) depends on behaviour at the top of the distribution, where responses are strong (Jakobsen et al. 2020; Kleven et al. 2014). Across the avoidance band the German dividend changes by −55 to +25 € per adult and month in 2028.

Take-up. The model assumes complete take-up of the floor. Today, non-take-up of minimum income is high in Germany and France (DREES 2022; Eurofound 2015); a payment without application and a pre-filled top-up are designed to reduce it, which raises costs that are already fully counted.

7 Assessment against the objectives

Table 11 relates each objective to the rules that serve it, to an indicator the model computes, and to what the model does not cover.

Table 11: Objectives, indicators and gaps

Objective	Rules that serve it	Indicator in the model (Germany unless stated)	Not covered
Tax fairness	one ceiling of 33 % for all kinds of income; no contributions on wages only; profits taxed where customers are	Gini 2025 → 2028: DE −4.0, FR −1.4, IT −4.1, ES −5.8, PL −6.2 points. Marginal burden at € 42,000 43.8 → 40.3 %; at € 100,000 53.1 → 39.1 %. Effective burden per euro: labour 48 → 21 %, capital 26 → 32 % (approximation)	base of company taxation where customers are; avoidance at the top only as a band
Intergenerational fairness	legacy debt and pension promises in a fund repaid by 2098; pension promises above basic need reduced; own funded account	fund 5.17 % of GDP until 2098, peak 148.1 % (2055); German pensioner households: mean −121 € per month, lowest fifth +24 €, highest fifth −478 €	no generational accounting by birth cohort; dividend path of later cohorts only as average
Climate	carbon price and 33 % rate on harmful goods returned as a dividend; lower VAT for certified goods	steering revenue 3.14 % of GDP (2028) → 0.67 % (2060); dividend 7.25 → 3.40 % of GDP	emissions are an exogenous path; no estimate of emission reductions caused by the rules
Digital transformation of work	social protection financed from all income, not from wages; dividend and net independent of employment; platforms taxed where customers are	shift of 3 GDP points from labour to capital income by 2060: revenue today −0.64 % of GDP, new system +0.33 %; dividend 2060 +10 % (Table 12)	scenario assumptions; platform revenue not quantified separately

Effective burden per euro: today tax wedge on labour (OECD) against flat capital income tax in Germany; new system revenue relative to the income base (dynamik33.toml, approximation). Marginal burden: single, Germany, without compulsory saving.

Tax fairness. Inequality of disposable income falls in all five countries. The burden on labour falls and the burden on capital rises, which is the intended shift. The marginal burden on high wages also falls, however: at € 100,000 from 53.1 % to 39.1 %, at € 250,000 from 44.3 % to 40.0 % (Table 7). The proposal thus gives up part of the progressivity on high wages in exchange for equal treatment of income types; whether this is fair depends on whether top incomes actually pay the 33 % on capital income, inheritance and profits, which the model treats with an avoidance band (section 6). One-earner couples lose the advantage of joint taxation; this is consistent with individual taxation but is a loss for these households.

Intergenerational fairness. Repayment by 2098 means that taxpayers after 2098 no longer carry today's legacy. The cost falls on taxpayers until then through the surcharge, and on pensioners with promises above basic need: German pensioner households lose 121 € per month on average, while the lowest fifth gains slightly. A full assessment would need generational accounts by birth cohort, which the model does not compute.

Climate: incentive instead of control. The rules are consistent with the objective in that harmful consumption is taxed and the revenue returned. Three elements work through prices rather than prohibitions:

- **Carbon price.** The European carbon price for buildings and transport (ETS2) enters as a revenue path; its revenue falls with emissions even though the price rises. Road transport fuel expenditure follows the German projection report 2026 (with-measures scenario).
- **Special rate on harmful goods.** Goods classified as harmful to health or the environment carry 33 % – in the model motor fuels, combustion-engine vehicles, tobacco and alcohol; chemical pesticides and mineral fertilisers are included by decision of 4 November 2026 (not yet in the calculation; the household share of these purchases is small). Because firms deduct VAT as input tax, the special rate does not reach agricultural use of pesticides and fertilisers; a levy differentiated by harmfulness, as in Denmark, is an open proposal. The model assumes that the share of combustion engines in vehicle purchases falls from 80 % in 2028 to 10 % in 2035 (assumption, oriented to EU vehicle CO_2 standards).
- **VAT label.** Table 13 shows the price signal implied by the rates alone. For the same net price, a certified product in class A costs 5.1 % less than a standard product; a harmful product costs 13.7 % more. A supplier who keeps the consumer price unchanged earns 5.4 % more per unit after moving from B to A. Certification pays if its cost is lower than this advantage.

Table 13: VAT label classes and the price signal

Class	Rate	Goods (examples)	Price vs B (1)	Revenue vs B (2)
A+	5 %	basic needs, repairs, circular products	−10.3 %	+11.4 %
A	11 %	certified sustainable goods	−5.1 %	+5.4 %
B	17 %	standard goods	0.0 %	0.0 %
C	25 %	resource-intensive goods	+6.8 %	−6.4 %
Special	33 %	harmful goods (e.g. motor fuels, combustion-engine vehicles, tobacco, alcohol, chemical pesticides, mineral fertilisers)	+13.7 %	−12.0 %

(1) Consumer price relative to class B at the same net price. (2) Supplier net revenue relative to class B at the same consumer price. Mechanical price signals from the rates alone. Which share of a rate difference reaches consumers or stays with suppliers depends on pass-through, which is asymmetric for VAT changes (Benzarti et al. 2020). Assignment of goods to classes is set by law; lower classes A+ and A are reached through certification.

The label is distributionally progressive in its burden (Table 6), and the revenue returns per head through the dividend. Evidence supports the direction of the mechanism but not its size: consumers respond more strongly to taxes shown at the point of purchase (Chetty, Looney and Kroft 2009); in a systematic review of environmental labels on

food, 60 of 76 comparisons favoured the labelled option, with many studies at unclear or high risk of bias (Potter et al. 2021). Certification works only if certifiers are independent and supervised (Dranove and Jin 2010). VAT cuts are passed on to consumers less than increases (Benzarti et al. 2020), so part of the advantage of class A may stay with suppliers – which still rewards certification, but weakens the price signal to consumers. Carbon rebates are often underestimated by recipients (Mildenberger et al. 2022).

The model does not estimate how far the rules reduce emissions: emissions are an exogenous path, and the demand response to the label is not quantified for the current rates. The legal frame is open. Current EU law allows at most two reduced rates of at least 5 % and one lower rate (Council Directive (EU) 2022/542), while the label has three rates below the standard class; rates that depend on a supplier's certification rather than on the product raise the question of fiscal neutrality (CJEU case C-515/20; secondary source). The same directive already points in the direction of the label: it permits reduced rates for solar panels and ends reduced rates for fossil fuels from 2030.

The success of climate policy reduces steering revenue from 3.14 % of GDP in 2028 to 0.67 % in 2060 and is the main reason why the dividend falls. Later cohorts therefore receive a smaller dividend – a tension with intergenerational fairness that the initiative accepts as intended.

Digital transformation of work. The clearest quantitative result concerns the tax base. Today a shift of three points of GDP from labour to capital income by 2060 would reduce revenue by 0.64 % of GDP, because capital income is taxed more lightly; in the new system the same shift raises revenue by 0.33 % of GDP. Faster productivity growth from automation raises the dividend in all scenarios (Table 12). Protection no longer depends on a job. Whether displacement by automation is large, and how labour markets adjust, is outside the model; the scenarios are assumptions.

Table 12: Dividend in 2060 under employment and automation scenarios

Scenario (2060)	Dividend vs baseline	Fund contribution (% GDP)
Baseline	0 %	5.17
Capital share +3 points	+10 %	5.17
AI productivity +0.07 pp/year	+3 %	5.13
AI productivity +0.3 pp/year	+14 %	5.00
AI productivity +0.6 pp/year	+31 %	4.83
Employment +2.8 %	+4 %	5.12
Employment +4.6 %, AI +0.3, capital +3	+33 %	4.92

dynamik33.json. AI range: Acemoglu (2024) to Filippucci, Gal and Schief (2024); employment from lower labour costs and lower marginal burden with assumed elasticities; capital share shift: assumption. Fund contribution as approximation.

8 Behavioural effects not in the model

A qualitative review of the literature (available in German in the research kit) assessed the expected responses to each building block:

- **Dividend:** experimental and quasi-experimental evidence finds small reductions in hours and little reduction in employment (Jones and Marinescu 2022; Vivalt et al. 2024; STM 2020).
- **Payment without application:** lowers non-take-up (Bhargava and Manoli 2015; Eurofound 2015); costs are already fully counted.
- **Compulsory saving with a default fund:** raises saving of passive savers, with partial crowd-out among active savers (Madrian and Shea 2001; Chetty et al. 2014).
- **Individual taxation:** raises hours of second earners (Bick and Fuchs-Schündeln 2018).

- **VAT label:** cuts reach consumers less than increases (Benzarti et al. 2020); consumers respond more strongly to taxes that are visible at the point of purchase (Chetty, Looney and Kroft 2009), which may strengthen the steering effect of a label (not tested).
- **Carbon price with rebate:** rebates are underestimated and do not reliably raise support (Douenne and Fabre 2022; Mildenberger et al. 2022).
- **Pension rule E:** loss aversion and resistance in the largest group of voters are an acceptance risk not captured by the model.

None of these findings questions the calculation as such; several show that outcomes depend on design details that are not yet fixed. A quantitative stage is open.

9 Limitations and open questions

- The model is static; behavioural responses are discussed but not quantified.
- Country models are simplified: one earner per household; regional rules and housing benefits beyond those named are not modelled; several parameters rest on secondary sources.
- The share of employees better off uses the German wage distribution scaled to each country's average wage; national wage distributions and two-earner couples are open.
- VAT burdens by income are available only for Germany and France. The demand response to the label and the effect of the rules on emissions are not quantified; the label requires changes to EU VAT law.
- The tax base of company profits taxed where customers are has not been checked against an independent source.
- The common fund of 30 states is scaled from ten states; interest and growth are assumptions with a wide band.
- Compatibility with EU law and national constitutions has not been assessed.
- Which states could adopt common rules is open. Broad acceptance and reliable enforcement are preconditions; the World Justice Project's Rule of Law Index (2025) is one possible criterion. States outside Europe are not calculated.

10 Reviews, corrections and access

The calculation chain was recalculated in three internal reviews in October 2026, initiated by the initiative and carried out with a different AI model from the one used to build the calculation; they are not independent reviews. The second and third reviews re-implemented the core in their own code and confirmed it at rounding level; their findings and the responses are documented. The last review covered model version v4.42; the changes since then (up to v4.70) are listed on the website; a follow-up review of them is under way, in which the fund grid (D1) and the marginal burden (D2) have been reproduced in full so far. Independent review is invited.

The log records every version with date, change, data source, moved figure and how it was noticed; 85 corrections have been numbered so far. The research kit with all figures to cite, the result files and the review documents is archived at doi.org/10.5281/zenodo.23855733. Scripts, parameters and the full log are public at github.com/archicrowd/2828.

References

1. Acemoglu, D. (2024): The Simple Macroeconomics of AI. NBER Working Paper 32487.
2. Benzarti, Y., Carloni, D., Harju, J., Kosonen, T. (2020): What Goes Up May Not Come Down: Asymmetric Incidence of Value-Added Taxes. Journal of Political Economy 128(12).

3. Bhargava, S., Manoli, D. (2015): Psychological Frictions and the Incomplete Take-Up of Social Benefits. American Economic Review 105(11).
4. Bick, A., Fuchs-Schündeln, N. (2018): Taxation and Labour Supply of Married Couples across Countries. Review of Economic Studies 85(3).
5. Chetty, R., Friedman, J., Leth-Petersen, S., Nielsen, T., Olsen, T. (2014): Active vs. Passive Decisions and Crowd-Out in Retirement Savings Accounts. Quarterly Journal of Economics 129(3).
6. Chetty, R., Looney, A., Kroft, K. (2009): Salience and Taxation: Theory and Evidence. American Economic Review 99(4).
7. Council of the European Union (2022): Council Directive (EU) 2022/542 of 5 April 2022 amending Directives 2006/112/EC and (EU) 2020/285 as regards rates of value added tax. Official Journal L 107.
8. Court of Justice of the European Union (2022): Case C-515/20, B & L Elektrotechnik (reduced VAT rates and the principle of fiscal neutrality); cited after a secondary source.
9. DREES (2022): Mesurer régulièrement le non-recours au RSA et à la prime d'activité. Les Dossiers de la DREES 92.
10. Dranove, D., Jin, G. Z. (2010): Quality Disclosure and Certification: Theory and Practice. Journal of Economic Literature 48(4).
11. Douenne, T., Fabre, A. (2022): Yellow Vests, Pessimistic Beliefs, and Carbon Tax Aversion. American Economic Journal: Economic Policy 14(1).
12. Eurofound (2015): Access to social benefits: Reducing non-take-up. Luxembourg: Publications Office of the European Union.
13. Eurostat: datasets gov_10a_exp, gov_10a_taxag, gov_10dd_edpt1, nama_10_gdp, nasa_10_pens1, demo_pjanind, une_rt_a, prc_ppp_ind (retrieved 17 October 2026); ilc_di12, ilc_di01 (EU-SILC 2025, retrieved 18 October 2026); hbs_str_t223 (Household Budget Survey 2020, retrieved 24 October 2026).
14. Filippucci, F., Gal, P., Schief, M. (2024): Miracle or myth? Assessing the macroeconomic productivity gains from artificial intelligence. OECD; summary on VoxEU, 8 December 2024.
15. Jakobsen, K., Jakobsen, K., Kleven, H., Zucman, G. (2020): Wealth Taxation and Wealth Accumulation: Theory and Evidence from Denmark. Quarterly Journal of Economics 135(1).
16. Jones, D., Marinescu, I. (2022): The Labor Market Impacts of Universal and Permanent Cash Transfers: Evidence from the Alaska Permanent Fund. American Economic Journal: Economic Policy 14(2).
17. Kleven, H., Landais, C., Saez, E., Schultz, E. (2014): Migration and Wage Effects of Taxing Top Earners. Quarterly Journal of Economics 129(1).
18. Madrian, B., Shea, D. (2001): The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior. Quarterly Journal of Economics 116(4).
19. Mildenberger, M., Lachapelle, E., Harrison, K., Stadelmann-Steffen, I. (2022): Limited impacts of carbon tax rebate programmes on public support for carbon pricing. Nature Climate Change 12(2).
20. OECD (2025): Taxing Wages 2025. Paris: OECD Publishing. – OECD (2026): Taxing Wages 2026. Paris: OECD Publishing.
21. Potter, C., Bastounis, A., Hartmann-Boyce, J., Stewart, C., Frie, K., Tudor, K., Bianchi, F., Cartwright, E., Cook, B., Rayner, M., Jebb, S. A. (2021): The Effects of Environmental Sustainability Labels on Selection, Purchase, and Consumption of Food and Drink Products: A Systematic Review. Environment and Behavior 53(8).
22. Saez, E., Schoefer, B., Seim, D. (2019): Payroll Taxes, Firm Behavior, and Rent Sharing: Evidence from a Young Workers' Tax Cut in Sweden. American Economic Review 109(5).
23. Statistisches Bundesamt (Destatis): Einkommens- und Verbrauchsstichprobe (EVS) 2023, Konsumausgaben privater Haushalte.
24. STM – Ministry of Social Affairs and Health, Finland (2020): Results of Finland's basic income experiment.
25. Umweltbundesamt et al. (2026): Projektionsbericht 2026 für Deutschland (with-measures scenario, transport).
26. Vivalt, E., Rhodes, E., Bartik, A., Broockman, D., Miller, S. (2024): The Employment Effects of a Guaranteed Income: Experimental Evidence from Two U.S. States. NBER Working Paper 32719.
27. World Justice Project (2025): Rule of Law Index 2025. Washington, DC.

Appendix: Figures to cite

No.	Item	Value	Unit	Status
Z01	Transformation dividend per adult (Germany)	**354**	€ per month	adopted
Z02	Transformation dividend per child (Germany)	**283**	€ per month	adopted
Z03	Gain, single, gross wage € 42,000 (Germany)	**611**	€ per month	adopted
Z04	Gain, couple with one earner and 2 children, € 42,000 (Germany)	**674**	€ per month	adopted
Z05	Working-age households better off (Germany)	**85.1**	%	check
Z06	All households better off (Germany)	**69.3**	%	check
Z07	Pensioner households better off (pension rule E)	**32.2**	%	adopted
Z08	Pensioner households, mean change (pension rule E)	**−121**	€ per month	adopted
Z09	Contribution to the common fund of 30 states	**5.17**	% of GDP	adopted
Z10	Legacy surcharge on all taxes (common rate)	**15.5**	%	adopted
Z11	Debt ratio of the fund, peak	**148.1**	% of GDP	adopted
Z12	Fund fully repaid	**2098**	year	adopted
Z13	Old pension claims above the dividend (Z)	**9.42**	% of GDP	adopted
Z14	Total burden, taxes and fund (Germany)	**38.7**	% of GDP	adopted
Z15	Total burden including compulsory saving (own account)	**42.8**	% of GDP	adopted
Z16	Cost of the strong net (Germany)	**9.63**	% of GDP	adopted
Z17	Marginal burden at € 60,000: today / new 2028 (Germany)	**48.8 / 40.0**	% per additional euro	check
Z18	VAT label burden, factor lowest / highest income class (Germany)	**0.87 / 1.07**	relative to mean	adopted
Z19	One-off price level effect of the VAT label 2028	**1.2 to 2.1**	%	check
Z20	Gain, single employee at the average wage (Germany)	**726**	€ per month	adopted
Z21	Gain, single employee at the average wage (France)	**276**	€ per month	adopted
Z22	Gain, single employee at the average wage (Italy)	**412**	€ per month	adopted
Z23	Gain, single employee at the average wage (Spain)	**321**	€ per month	adopted
Z24	Gain, single employee at the average wage (Poland)	**1,404**	zł per month	adopted
Z25	Legacy surcharge 2028 after netting (Germany)	**15**	%	adopted
Z26	Legacy surcharge 2028 after netting (France)	**21.3**	%	adopted
Z27	Legacy surcharge 2028 after netting (Italy)	**22.3**	%	adopted
Z28	Legacy surcharge 2028 after netting (Spain)	**18.3**	%	adopted
Z29	Legacy surcharge 2028 after netting (Poland)	**13.5**	%	adopted
Z30	Gini coefficient 2025 → 2028 (Germany)	**30.1 → 26.1**	index 0–100	check
Z31	Gini coefficient 2025 → 2028 (France)	**30.4 → 29.0**	index 0–100	check
Z32	Gini coefficient 2025 → 2028 (Italy)	**31.0 → 26.9**	index 0–100	check
Z33	Gini coefficient 2025 → 2028 (Spain)	**30.8 → 25.0**	index 0–100	check
Z34	Gini coefficient 2025 → 2028 (Poland)	**24.9 → 18.7**	index 0–100	check

Source file and path of each figure: figures-to-cite.csv in the research kit (DOI 10.5281/zenodo.23855733). adopted: under the decisions of the initiative; check: cross-check or sensitivity.